

Database security

MariaDB is a community-developed, commercially supported [fork](#) of the [MySQL relational database management system](#) (RDBMS), intended to remain [free and open-source software](#) under the [GNU General Public License](#). Development is led by some of the original developers of MySQL, who forked it due to concerns over its [acquisition](#) by [Oracle Corporation](#) in 2009.

`sudo apt install mariadb-server -y`

`sudo systemctl start mariadb` [starts mariadb services]

`sudo systemctl enable mariadb` [Makes MariaDB start **automatically on boot**]

`sudo mysql_secure_installation` [interactive shell thaa...many qns]

- 1. Switch to unix_socket authentication? Root login works only from system (not password-based)
- 2. Set root password?
- 3. Remove anonymous users? Removes users without login name (security risk)
- 4. Disallow root login remotely?Y
Prevents root login from outside machine
- 5. Remove test database? Test DB is useless + insecure
- 6. Reload privilege tables? Applies all changes immediately

`sudo mysql` - Opens MariaDB shell as root

```
hekh@hekh:~$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 39
Server version: 10.6.23-MariaDB-0ubuntu0.22.04.1 Ubuntu 22.04

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE secure_onboarding;
Query OK, 1 row affected (0.009 sec)

MariaDB [(none)]> CREATE USER 'secure_user'@'localhost' IDENTIFIED BY '1234';
Query OK, 0 rows affected (0.006 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON secure_onboarding.* TO 'secure_user'@'localhost';
Query OK, 0 rows affected (0.005 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT
Bye
```

I created a db named `secure_onboarding`, then created a user `secure_user` (that runs only on same machine) with password 1234, and granted all access to this db tables (wildcard). Then flushed privileges so that changes apply immediately (reload).

`sudo nano /etc/mysql/mariadb.conf.d/50-server.cnf` (config file)

`bind-address = 127.0.0.1`

127.0.0.1 is a special IP address

It always means: “this computer itself” (loopback address)

Binding = attaching a service to an IP address + port

(Only programs running inside the same VM can connect–Maria db is only accessible from this vm)

Backup setup:

`sudo nano /usr/local/bin/db_backup.sh`

Inside that : `#!/bin/bash`

`mysqldump -u root secure_onboarding > /var/backups/secure_onboarding.sql`

mysqldump → tool to export database

-u root → username = root

secure_onboarding → DB name

> → redirect output

/var/backups/... → where backup is saved

`sudo chmod +x /usr/local/bin/db_backup.sh` (change permissions,+x=add execution)

Now : scheduling - `sudo crontab -e`

```
# m h dom mon dow  command
0 2 * * * /usr/local/bin/backup_exam.sh
0 3 * * * /usr/local/bin/db_backup.sh|
```

Added this at 3:00 daily

```
MariaDB [(none)]> ALTER USER 'secure_user'@'localhost' IDENTIFIED BY '123456';
Query OK, 0 rows affected (0.003 sec)
```

To change password;)